

Noel, Jared L.
COM243-ML

2. Compute Distances (10 points)

H4							
	A	B	C	D	E	F	G
1	Student	Hours Studied (X)	Prep Hours (X)	Pass/Fail (Y)	clidean Distance		
2	1	1	5	0	3.16227766		
3	2	2	5.5	0	2.061552813		
4	3	3	6	0	1		
5	4	4.5	5	0	1.118033989		
6	5	5	6.5	1	1.118033989		
7	6	5.5	7	1	1.802775638		
8	7	6	6	1	2		
9	8	7	7	1	3.16227766		
10	9	8	6	1	4		
11	10	9	7.5	1	5.220153254		
12							
13							
14							
15							

2. Find the 3 Nearest Neighbors (3 points)

	Student	Hours Studied (X)	Prep Hours (X)	Pass/Fail (Y)	clidean Distance	
	1	1	5	0	3.16227766	
	2	2	5.5	0	2.061552813	
	3	3	6	0	1	
	4	4.5	5	0	1.118033989	
	5	5	6.5	1	1.118033989	
	6	5.5	7	1	1.802775638	
	7	6	6	1	2	
	8	7	7	1	3.16227766	
	9	8	6	1	4	
	10	9	7.5	1	5.220153254	

3. Majority Vote (2 points)

- Count how many are Pass (1) and how many are Fail (0).
- Predict the outcome for the new student.

Student	Hours Studied (X1)	Sleep Hours (X2)	Pass/Fail (Y)	Euclidean Distance
1	1.0	5.0	0	?
2	2.0	5.5	0	?
3	3.0	6.0	0	?
4	4.5	5.0	0	?
5	5.0	6.5	1	?
6	5.5	7.0	1	?
7	6.0	6.0	1	?
8	7.0	7.0	1	?
9	8.0	6.0	1	?
10	9.0	7.5	1	?

Prediction: A new student who studied 4 hours and slept 6 hours will _____ the exam.

4. Discussion Questions (5 points)

What was your final prediction?

My final prediction for the new student who studied for 4 hours and slept for 6 hours will 0 or fail the exam.

How would the prediction change if we used $k = 5$ instead of $k = 3$?

If we use $k = 5$ instead of $k = 3$, the result would be different and the new student will pass the exam.

Based on the data, there would be more than 1 neighbor and the prediction would result in the new student passing.

0	1
0	1.118033989
1	1.118033989
1	1.802775638
1	2